

Product Requirements Document (PRD)

Earnings Volatility (IV Crush) Options Trading Application • Version 3.0

1. Product Overview

This application automates the detection, structuring, and position sizing for an earnings-based options trading strategy. By capitalizing on implied volatility (IV) crush, the app recommends **Long Calendar Spreads** on equities approaching earnings announcements. It relies on strict statistical edges and rigorous risk management derived from backtesting 72,500 historical earnings events.

2. Data Ingestion & Metric Calculation

The backend must interface with a financial data provider to pull historical and real-time options chain data. The engine must compute three specific variables per ticker:

- **Term Structure Slope:** $Slope = IV_{near} - IV_{45+}$. We require backwardation (negative slope), meaning short-term certainty is being overbought.
- **30-Day Average Volume:** Moving average of the underlying equity's daily volume over the last 30 trading days to guarantee liquidity.
- **IV30/RV30 Ratio:** $Ratio = IV_{30} / RV_{30}$. Compares forward-looking 30-day implied volatility to backward-looking 30-day realized volatility.

3. Recommendation Engine Logic

The filtering system applies deterministic rules to the metrics to output a specific trading signal.

Signal	Logic Conditions
Recommend	$Slope < 0$ AND $Volume > Threshold$ AND $IV30/RV30 > Threshold$
Consider	$Slope < 0$ AND ($Volume \leq Threshold$ OR $IV30/RV30 \leq Threshold$)
Avoid	$Slope \geq 0$ (Term structure is in contango or flat)

4. Trade Structuring: Long Calendar Spread

For tickers labeled **Recommend**, the system must automatically build the trade block:

- **Leg 1 (Short):** Sell the At-The-Money (ATM) Call/Put expiring nearest to the earnings date.
- **Leg 2 (Long):** Buy the At-The-Money (ATM) Call/Put (same strike) expiring approximately 30 days after Leg 1.

5. Risk Management: The 10% Kelly Criterion

Position sizing is the most critical component of this strategy. The traditional Kelly formula determines the optimal fraction (f^*) of a bankroll to wager to maximize logarithmic wealth growth:

$$f^* = p - (q / b)$$

To smooth the equity curve and protect against black swan earnings reactions, a fractional multiplier is hardcoded:

$$f_{\text{applied}} = 0.10 \times f^* = 0.06$$

The app enforces a strict limit: $\text{Max Debit Allocation} = \text{Total Portfolio Value} \times 0.06$.

6. Advanced Quantitative Optimizations (v3.0 Upgrades)

To isolate the absolute highest-conviction trading setups and mitigate directional exposure, the recommendation and execution engines are augmented with four advanced statistical filters.

6.1 IV Percentile / Rank Filter

The system must track the underlying asset's Implied Volatility over a 252-trading-day rolling window to calculate the **IV Percentile**:

$$\text{IV Percentile} = \frac{\text{Days where IV}_{\text{current}} > \text{IV}_{\text{historical}}}{252} \times 100$$

- **Rule Constraint:** Modify the "Recommend" signal gate to require $\text{IV Percentile} \geq 70\%$.
- **Purpose:** Ensures options are priced at a true historical extreme relative to the stock's native behavior, maximizing the velocity of the next-day IV Crush.

6.2 Earnings Magnitude Filter (Expected vs. Historical Realized Move)

The backend must compute the options market's **Expected Move (EM)** using the front-week At-The-Money Straddle price:

$$EM \approx 0.85 \times (\text{Price}_{\{ATM\} \text{ Call}} + \text{Price}_{\{ATM\} \text{ Put}})$$

The engine will query historical data to calculate the mean absolute percentage move over the last 4 to 8 quarters ($\text{Move}_{\{hist\}}$).

- **Rule Constraint:** Prioritize tickers where $EM > \text{Move}_{\{hist\}} \times 1.25$ (Market is pricing a move 25%+ wider than historical tendencies).
- **Purpose:** Exploits over-hedging and extreme fear premium baked into option pricing by institutional market participants.

6.3 Multi-Strike Vega Optimization

When an asymmetrical directional drift is detected in historical quarterly performance data, the execution engine must evaluate slightly Out-of-the-Money (OTM) strikes.

- **Logic:** If a stock historically moves upward on 80% of Q2 cycles, evaluate a strike placed $+0.5 \times EM$ above current spot price.
- **Purpose:** Lowers the initial debit paid (reducing absolute maximum risk) while engineering a high-yielding performance zone if the asset lands near the selected strike.

6.4 Dynamic Exit Protocols

The software must implement strict programmatic exit thresholds instead of relying on an arbitrary time-based exit.

- **Take-Profit Threshold:** Automatically trigger an exit alert if the spread value achieves a $+25\% \text{ ext}\{to\} +35\%$ return on the premium paid.
- **Velocity Exit:** Monitor the front-week contract's IV during the first 5 minutes of the opening bell. If $\geq 80\%$ of the expected IV collapse occurs immediately, trigger an aggressive sell signal to lock in gains before directional market trends emerge.

7. Strategy Validation: Monte Carlo Simulation Module

Simulates 1,000 parallel paths across 500 trades using historical win rates (62%) and sizing constraints (6%) to plot 5th, 50th, and 95th percentile equity curves and evaluate the Risk of Ruin.

8. UI / UX Requirements

Streamlit implementation featuring dashboard inputs, visual signal banners, metric dataframes, and specific position-size readouts in dollar amounts.